

HKD-Net: Hierarchical Knowledge Distillation Based on Multi-Domain Feature Fusion for Efficient Network Intrusion Detection

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Abstract—We propose HKD-Net, a hierarchical knowledge distillation network based on multi-domain feature fusion, for efficient network intrusion detection on resource-constrained edge devices. The framework incorporates dedicated feature extraction modules across temporal, frequency, and spatial domains, and introduces a dynamic gating mechanism for adaptive feature fusion, resulting in a more discriminative and comprehensive feature representation. Moreover, a hierarchical distillation mechanism is designed that not only preserves soft labels from the output layer but also aligns intermediate features from spatial, temporal, frequency, and fused domains, enabling efficient knowledge transfer from a large teacher model to a compact student model. Through knowledge distillation, the final lightweight model requires only 278,580 parameters, reducing the number of parameters by approximately 74.68% compared to the teacher, while maintaining high detection accuracy. Extensive experiments on three public datasets (Kitsune, CIRA-CIC-DoHBrw2020, and CICIoT2023) demonstrate that HKD-Net outperforms five state-of-the-art methods, achieving accuracies of 96.72%, 97.19%, and 87.19%, respectively, while reducing parameters by 74.68% and maintaining low computational cost.

Index Terms—Network traffic anomaly detection, knowledge distillation, multi-domain feature, deep learning, network intrusion detection.

I. INTRODUCTION

WITH the rapid development of network technology and the increasing complexity of network attack methods, network intrusions have become increasingly covert, frequent, and advanced [1]. In recent years, organized and targeted cyberattacks, such as data breaches, cyberfraud, and ransomware, have become frequent, posing severe challenges to cybersecurity [2]. To address these threats, network intrusion

detection (NID) technology has emerged, aiming to detect and identify malicious activities within networks to provide robust security safeguards for cyberspace [3], [4], [5]. However, NID systems face significant challenges in balancing detection accuracy with computational efficiency, especially on resource-constrained edge devices.

Compared to traditional machine learning methods, deep learning demonstrates powerful automatic feature extraction capabilities when handling large-scale intrusion traffic, making it widely adopted in the field of NID. Currently, researchers have employed various deep learning models to study NID, including convolutional neural networks (CNN) [6], long short-term memory networks (LSTM) [7], and multi-scale temporal convolutional networks (TCN) [8], among others. In the temporal dimension, since the introduction of recurrent neural network (RNN) models, they have been widely applied in the field of sequential data processing [9]. However, time-domain-based methods often fail to effectively utilize frequency characteristics and are prone to interference.

In the frequency domain, existing studies typically apply spectral or time-frequency transformations—such as the 2D Fourier transform [10], the combined fast Fourier transform and continuous wavelet transform [11], and the discrete wavelet transform (DWT) integrated into Transformer architectures [12] to convert time-series traffic data into frequency representations. These methods leverage the concentrated nature of low-frequency components and the distributed characteristics of high-frequency bands to enhance feature continuity and discriminability, thereby improving the performance of intrusion detection models. In the spatial domain, extracting spatial features using CNN has become a cutting-edge method [13]. Regarding feature fusion, Yao et al. [14] adopted the cross-layer feature fusion of the CNN and LSTM to separately obtain global and periodic features.

While recent efforts have integrated multi-domain features or adopted knowledge distillation (KD) for lightweight intrusion detection, they still exhibit notable limitations. For instance, methods like Dual-Modal Intrusion Detection System (DM-IDS) [19] fuse dual-modal features via bilinear attention but employ static fusion strategies, lacking adaptability to diverse attack patterns. TFTKD [10] leverages frequency-domain transformation and output-layer distillation yet overlooks intermediate feature representations, restricting knowledge transfer depth. Similarly, MT-Security [20]

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combines raw-byte and graph features but relies on complex architectures unsuitable for edge deployment.

To address these problems, we propose a hierarchical knowledge distillation network based on multi-domain feature fusion, termed HKD-Net. The framework adopts a hierarchical distillation strategy that aligns features from both the final output and intermediate layers across the spatial, temporal, and frequency domains. On this basis, a dynamic gating mechanism is introduced for adaptive fusion of cross domain features, thereby constructing a more discriminative and comprehensive feature representation. This fine-grained alignment ensures that the student model learns the rich semantic, structural, and contextual information embedded in the teacher's intermediate features, rather than merely mimicking its final decisions. Furthermore, by employing a dynamic balancing mechanism based on learnable weights, the training process automatically adjusts the relative importance of alignment losses at different levels, which achieves self-optimization of distillation. This method significantly enhances distillation performance while maintaining a lightweight architecture, offering a lightweight and efficient solution for resource-constrained environments. In summary, this paper's main contributions are as follows:

- HKD-Net is proposed to overcome the limitations of traditional single-feature extraction methods. The model can extract the spatial-temporal-frequency stream features in parallel and adaptively weight the contributions of different modalities based on attack characteristics, which constructs a more discriminative and comprehensive feature space.
- The hierarchical knowledge distillation framework systematically transfers the intermediate features of the teacher model to the student model by establishing alignment at multiple intermediate semantic levels, such as spatial, temporal, frequency, and fusion features, while maintaining the soft labels of the output layer. This approach enables a deeper and more structured form of knowledge transfer.
- The dynamic balancing mechanism based on learnable weights can adaptively calibrate the contribution of multi-level distillation losses according to training dynamics. Compared to static weight allocation, this strategy addresses the rigidity of fixed weighting schemes by dynamically adjusting the weights of each feature modality during the optimization process.

The rest of the paper is organized as follows: Section II reviews related work on deep learning-based intrusion detection and knowledge distillation. Section III presents the proposed algorithms. Section IV presents the analysis and results of the experiments conducted. In Section V, we discuss the strengths and limitations, and present future directions of our work. Last, we summarize our work in Section VI.

II. RELATED WORK

In this section, we provide a concise review of the relevant literature on network traffic anomaly detection from two methodological perspectives: deep learning approaches and knowledge distillation techniques.

A. DL-Based Traffic Anomaly Detection Methods

Deep learning technology has demonstrated significant advantages in handling large-scale intrusion traffic due to its powerful automatic feature extraction capabilities and has therefore been widely applied in the NID field [15]. Current research trends indicate that multi-domain feature extraction methods are shifting from single-feature analysis toward multi-domain collaboration to more comprehensively capture abnormal patterns in network traffic [16]. For instance, Xu et al. [17] introduced a channel attention mechanism to dynamically weight spatio-temporal-frequency features in industrial anomaly detection. Kotak et al. [18] employed V-NTC (VPN-encrypted Network Traffic Classification), a sliding-window time-series approach with InceptionTime, to classify VPN-encrypted traffic at both category and application levels. Zha et al. [19] proposed DM-IDS, a dual-modal attention-CNN architecture that fuses flow and binary-encoded payload features via bilinear fusion. Yang et al. [20] introduced MT-Security, which fuses raw-byte and graph-based interaction features to classify encrypted malicious traffic. D'Angelo et al. [22] employed CRAE-Net (Convolutional-Recurrent AutoEncoder Network), a stacked CNN-LSTM sparse-autoencoder architecture, to distill spatial-temporal features from raw traffic statistics. While these methods improve accuracy through complex multi-domain feature extraction, numerous rely on serial multi-network architectures that increase computational cost and memory consumption, limiting their applicability in resource-constrained environments.

B. Knowledge Distillation for Lightweight NID

Deep learning-based light weight methods have been applied to NIDSS. Ayyat et al. [21] present ClassyNet, a class-aware early-exit network that prioritizes high-priority threats for low-latency edge-side detection. Das et al. [23] propose a lightweight incremental IDS combining OCSVM with adaptive random forest to enable streaming detection and real-time blocking in underwater sensor networks. The aforementioned deep learning methods adopted lightweight structures, while the limited representation ability restricts the accuracy performance and makes the models sustain the risk of poor generalization ability.

Recently, Knowledge distillation (KD) is an effective technique for compressing large models by transferring knowledge from a teacher to a lightweight student model while preserving performance. Traditional response-based KD uses soft labels from the teacher's output layer to supervise the student via KL divergence. Zhao et al. [24] first introduced KD into NID, proposing a lightweight model using separable convolutions. Wang et al. [25] combined a triple-state CNN with K-fold cross-training, though such methods often lose structural and semantic information from intermediate layers, limiting generalization in fine-grained detection tasks. Feature-based KD addresses this by aligning intermediate feature representations between teacher and student. Wang et al. [26] used a spatio-temporal residual network to transfer feature extraction capabilities. Ding et al. [27] incorporated knowledge alignment and correlation modules to improve structured knowledge

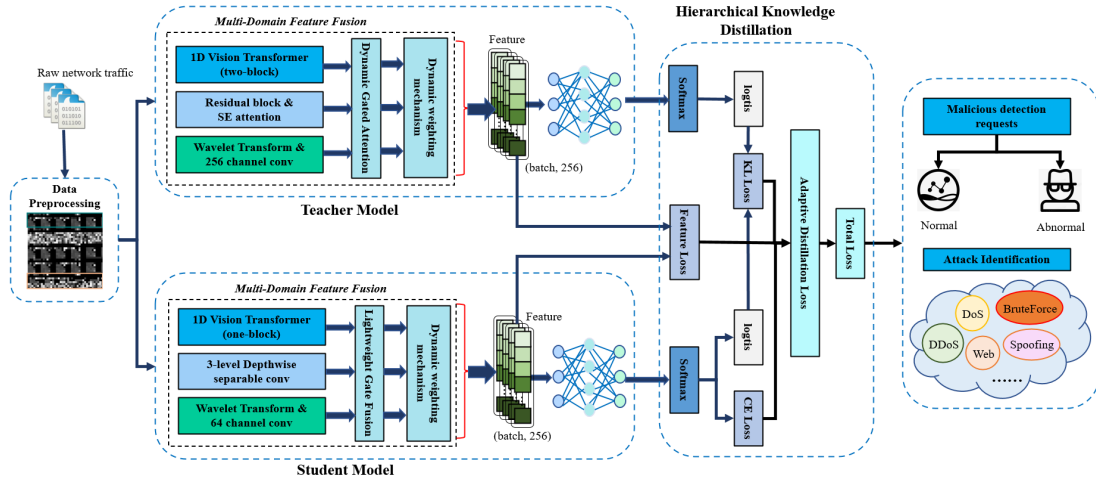


Fig. 1. Architecture of HKD-Net. The teacher model extracts and fuses multi-domain (spatial, temporal, frequency) features, the student model learns through hierarchical distillation, and the hierarchical distillation module aligns both output logits and intermediate features across domains to achieve efficient knowledge transfer.

transfer. AL-Nomasy et al. [28] built a Transformer-based teacher with selective gradient distillation to transfer spatio-temporal context features for IoT traffic detection.

Despite these advances, efficient transfer of multi-level features and maintaining high accuracy in lightweight models remain challenging. Our work addresses these limitations by proposing a hierarchical knowledge distillation network (HKD-Net) with multi-domain feature fusion, enabling efficient and accurate lightweight network intrusion detection.

III. METHODS

A. Network Traffic Anomaly Detection Framework

In this section, we propose HKD-Net, a lightweight network traffic anomaly detection framework based on hierarchical knowledge distillation with multi-domain feature fusion. As shown in Figure 1, we construct a teacher model consisting of a three-modal feature extraction module (1D-ViT time domain encoder, SE-ResNet spatial encoder, and wavelet frequency domain encoder), as well as a dynamic channel attention gating. This design enables high-accuracy joint optimization across domains. The lightweight student model employs depth-wise separable convolutions and a simplified ViT, and learns via a hierarchical distillation strategy. By synergistically transferring multi-modal knowledge and dynamically weighting feature alignments, HKD-Net effectively balances detection accuracy with computational efficiency.

B. Data Preprocessing

Inspired by the Multi-level Flow Representation (MFR) design in YaTC [29] we adopt a structured matrix-based approach for data preprocessing. First, raw network traffic is partitioned into individual flows based on IP addresses, port numbers, and protocol types to preserve flow-level context. To eliminate potential biases, we standardize the traffic by removing Ethernet headers, resetting port numbers to zero,

and replacing IP addresses with random values while preserving their directionality. Subsequently, we extract M adjacent packets from each flow and format them into an $H \times W$ two-dimensional matrix as the flow representation.

The matrix directly encodes raw byte values while adhering to a hierarchical structure:

- Byte-level: Each row contains only one type of traffic bytes (header or payload).
- Packet-level: Each packet is represented by its header and payload sub-matrices.
- Flow-level: M packet-level matrices are sequentially stacked to form the final MFR matrix.

By leveraging this preprocessing method, we effectively capture hierarchical traffic characteristics, providing high-quality input for subsequent multi-domain feature fusion and knowledge distillation.

C. Teacher Model

The teacher model is a core component of the multi-domain feature fusion network intrusion detection framework proposed in this paper. The objective of its design is to construct a high-performance feature representation system by integrating feature extraction modules from the temporal, spatial, and frequency domains. The model consists of four key modules: a temporal domain feature extraction module, a spatial-domain feature extraction module, a frequency-domain feature extraction module, and a dynamic feature fusion module. These modules collaborate to extract and fuse multi-domain features of network traffic data, providing high-quality feature representations for subsequent knowledge distillation.

1) *Temporal Domain Feature Extraction*: The temporal feature extraction module captures time series characteristics in network traffic data. This is crucial for identifying time-dependent attack patterns. Aiming at the long time-series dependency characteristics of network traffic data, this paper designs an improved one-dimensional Vision Transformer (1D-ViT) architecture, as shown in Figure 2 The inputs of

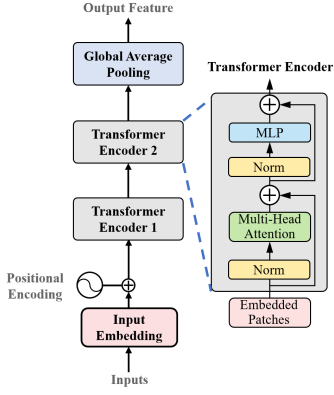


Fig. 2. Temporal feature extraction module based on a lightweight 1D Vision Transformer. The module processes sequential traffic data with multi-head self-attention and outputs a 64-dimensional temporal feature vector.

this module are time series numbers according to $\mathbf{T} \in \mathbb{R}^{n \times d}$, where n denotes the time step and d is the feature dimension. The original features are mapped to the embedding space through a linear projection layer:

$$\mathbf{E} = \mathbf{X}\mathbf{W}_e + \mathbf{P} \quad (1)$$

where $\mathbf{W}_e \in \mathbb{R}^{d \times d_{model}}$ is the learnable projection matrix ($d_{model} = 64$) and $\mathbf{P} \in \mathbb{R}^{n \times d_{model}}$ is the position encoding matrix whose element values are generated by the following equation:

$$E_{pos}(i, 2j) = \sin\left(\frac{i}{10000^{2j/d}}\right), \quad (2)$$

$$E_{pos}(i, 2j+1) = \cos\left(\frac{i}{10000^{2j/d}}\right) \quad (3)$$

The relative positional relationship of timing information can be effectively preserved by this positional encoding method.

The multi-head attention mechanism is the core component of the Transformer model's encoder. Its main purpose is to find relevant information in the input sequence and perform weighted summarization to produce the output sequence. Each group of attention mechanisms is called a "head", and each head learns a different representation and concatenates and projects these representations to obtain the final representation. In this paper, we use a lightweight multi-head attention structure as follows:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)\mathbf{W}^o \quad (4)$$

$$\text{head}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V) \quad (5)$$

where the number of attention heads $h = 4$, much less than the standard ViT configuration (usually 8-12 heads), and the dimension of each head $d_k = 16$. With this streamlined design, the computational complexity is reduced to $\mathcal{O}(4nd^2)$. The module finally performs feature abstraction through two tandem Transformer blocks and uses global average pooling to generate a 64-dimensional temporal domain feature vector $\mathbf{F}_t \in \mathbb{R}^{64}$.

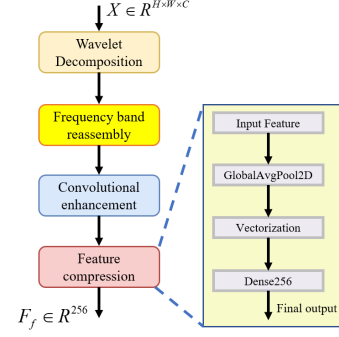


Fig. 3. Frequency domain feature extraction module utilizing Discrete Wavelet Transform with Daubechies-4 basis. The module decomposes traffic signals into low-frequency (LL) and high-frequency (LH, HL, HH) subbands, followed by energy normalization and convolutional processing.

2) *Spatial Domain Feature Extraction*: The spatial domain feature extraction module employs an improved SE-ResNet architecture to capture local spatial features from network traffic data, such as protocol fields and packet length statistics. This variant of ResNet integrates parallel multi-scale convolution with a channel attention mechanism (SE module) to mitigate the gradient vanishing problem through residual connections, thereby yielding more refined feature representations. A multi-scale convolutional branch is first constructed:

$$\mathbf{F}_{multi} = [\text{Conv}_{3 \times 1}(\mathbf{X}); \text{Conv}_{5 \times 1}(\mathbf{X}); \text{Conv}_{7 \times 1}(\mathbf{X})] \quad (6)$$

A channel attention mechanism is then incorporated to dynamically adjust channel weights via the Squeeze-and-Excitation (SE) module:

$$\mathbf{z}_c = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W u_c(i, j) \quad (7)$$

$$\mathbf{s} = \sigma(W_2 \delta(W_1 \mathbf{z})) \quad (8)$$

where δ denotes the ReLU activation function, σ is the Sigmoid function, and the compression ratio $r = 16$.

The module adopts a three-level residual structure. The final output is a 128-dimensional spatial feature vector $\mathbf{F}_s \in \mathbb{R}^{128}$, which contains complete spatial information from low-level protocol features to high-level behavioral patterns.

3) *Frequency Domain Feature Extraction*: The frequency domain feature extraction module aims to capture potential periodic patterns and sudden anomalies in network traffic, which are often difficult to identify effectively in time and space domain analysis. This module combines Discrete Wavelet Transform (DWT) with deep learning methods to realize multi-resolution frequency domain analysis. Figure 3 shows the architecture of the frequency domain feature extraction module.

The Daubechies-4 wavelet basis function is first used to decompose the input signal at multiple levels, the wavelet decomposition submodule:

$$[LL, (LH, HL, HH)] = \text{pywt.dwt2}(x, 'db4') \quad (9)$$

Specifically, the input signal is decomposed into multiple subbands using the Daubechies-4 wavelet basis function, which yields one low-frequency approximation component

(LL), which reflects the overall trend of the flow, and three high-frequency detail components (LH, HL, HH), which effectively capture abrupt traffic changes and anomalous pulses. Furthermore, a wavelet coefficient energy normalization layer is introduced:

$$\hat{C}_i = \frac{C_i}{\sqrt{\sum_{j=1}^N C_j^2}} \quad (10)$$

This layer effectively eliminates the interference caused by amplitude differences across traffic samples, thereby enhancing the model's cross-scenario generalization capability. The module ultimately outputs a 256-dimensional frequency domain feature vector, which, together with the spatial and temporal features, constitutes a comprehensive multi-domain representation system.

4) *Dynamic Feature Fusion*: To effectively integrate multi-domain features from the temporal, spatial, and frequency domains, HKD-Net employs a dynamic feature fusion strategy based on a gating mechanism. This strategy enables the model to adaptively select and combine features from different domains, enhancing its ability to characterize complex network traffic patterns.

Firstly, the features are projected to a uniform dimension by 1×1 convolution:

$$\hat{F}_k = W_k F_k, \quad k \in \{t, s, f\} \quad (11)$$

where $W_k \in \mathbb{R}^{d_k \times 256}$ is the projection matrix ensuring all features are in the same dimension space. The gated weight network uses a two-layer fully connected structure:

$$g = \delta \left(W_2 \cdot \delta \left(W_1 \left[\hat{F}_t; \hat{F}_s; \hat{F}_f \right] \right) \right) \quad (12)$$

$$w_k = \frac{e^{g_k}}{\sum_{j \in \{t, s, f\}} e^{g_j}} \quad (13)$$

where δ is the ReLU activation function, and W_1 and W_2 are weight matrices. The final fusion feature is obtained by a weighted sum and residual connection:

$$F_{\text{fused}} = \sum_{k \in \{t, s, f\}} w_k \hat{F}_k + W_r \left[\hat{F}_k; \hat{F}_s; \hat{F}_f \right] \quad (14)$$

$W_r \in \mathbb{R}^{768 \times 256}$ is the residual projection matrix. This structure preserves the distinctiveness of each modality while enhancing inter-feature interactions.

D. Student Model

The student model consists of a lightweight and efficient spatiotemporal network structure, which integrates multi-domain information through deep separable convolution, simplified Transformer architecture, and efficient wavelet processing, and finally through a dynamic gating fusion module.

1) *Lightweight Feature Extraction*: The student model employs a lightweight and efficient spatiotemporal network structure, integrating multi-domain information through deep separable convolution, simplified Transformer architecture, and efficient wavelet processing. In the spatial domain branch, Depthwise Separable Convolution is used instead of standard convolution. This technique decomposes traditional convolution operations into depthwise and pointwise convolution,

significantly reducing computational complexity while retaining the feature extraction capabilities of convolutional neural networks. The processing flow is as follows:

$$\text{DepthwiseConv}(X) = \text{Depthwise}(X) * W_{\text{depthwise}} + b, \quad (15)$$

$$\text{Pointwise}(X) = \text{Conv}_{1 \times 1}(X) * W_{\text{pointwise}} \quad (16)$$

this design reduces the number of parameters of standard convolution from $\mathcal{O}(k^2 \times C_{in} \times C_{out})$ to $\mathcal{O}(k^2 \times C_{in} + C_{in} \times C_{out})$, which significantly reduces the model complexity.

The 1D Vision Transformer (ViT) in the temporal domain retains only a single-layer Transformer block and 2 attention heads, using global average pooling to replace the fully connected layer. This results in a 64-dimensional time-domain feature vector $\mathbf{F}_{\text{temporal}} \in \mathbb{R}^{64}$, reducing computation by more than 60%. The frequency domain processing branch uses a truncated wavelet transform strategy, performing one-level Daubechies-1 wavelet decomposition on the input signal and preserving the low-frequency LL subbands. The subband coefficients are processed by a 3×3 convolution kernel with shared weights, and the 64-dimensional frequency domain features are generated by global pooling. This lightweight design ensures that the student model can be efficiently deployed on resource-constrained edge devices.

2) *Efficient Fusion Module*: The student model's fusion module employs a dynamic gating mechanism to integrate multi-domain features adaptively. Given spatial domain features $\mathbf{f}_s \in \mathbb{R}^{d_s}$, temporal features $\mathbf{f}_t \in \mathbb{R}^{d_t}$ and frequency domain features $\mathbf{f}_f \in \mathbb{R}^{d_f}$, they are first projected to a unified dimension using linear transformations:

$$\tilde{f}_i = W_i f_i + b_i, \quad i \in \{s, t, f\} \quad (17)$$

where $W_i \in \mathbb{R}^{d_u \times d_i}$ are learnable projection matrices. The gating network then calculates modality weights:

$$w = \text{Softmax}(W_2 \delta(W_1 [\tilde{f}_s; \tilde{f}_t; \tilde{f}_f] + b1) + b2) \quad (18)$$

where $W_1 \in \mathbb{R}^{3d_u \times d_m}$ and $W_2 \in \mathbb{R}^{d_m \times 3}$ are weight matrices, and δ is the ReLU activation function. The final fused features are obtained by:

$$f_{\text{fused}} = \text{LayerNorm} \left(\sum_{i \in \{s, t, f\}} w_i \tilde{f}_i + W_r [f_s; f_t; f_f] \right) \quad (19)$$

where $W_r [f_s; f_t; f_f]$ are used to supplement the original feature interaction information. This design not only maintains the discriminative combination of multi-domain features, but also enhances the robustness of features through cross-modal connections.

E. Hierarchical Knowledge Distillation

Hierarchical knowledge distillation is an effective model compression technique designed to migrate knowledge from the teacher model to the lightweight student model. The framework consists of three core components: softened-label distillation at the output layer, alignment of the intermediate feature layer, and adaptive loss weighting, which form an end-to-end knowledge learning to ensure that the student model

maintains performance close to that of the teacher model while significantly reducing computational complexity.

1) *Output Layer Distillation (Softened Labels)*: Using the standard knowledge distillation loss of the output layer, this loss utilizes the softmax of the temperature scale to generate softening labels from the teacher model. The student model is trained to mimic this softened output distribution, thereby inheriting the teacher's inter-class relational knowledge. The distillation loss is defined by the Kullback-Leibler (KL) divergence:

$$L_{KL} = \sum_{i=1}^C p_i^{(s)} \log \left(\frac{p_i^{(s)}}{p_i^{(t)}} \right) \quad (20)$$

where $p_i^{(s)}$ and $p_i^{(t)}$ represent the temperature-scaled softmax probabilities of the student and teacher models for the i -th class, respectively, and C is the total number of categories. The temperature parameter $T > 1$ smooths the probability distribution, facilitating more effective knowledge transfer.

2) *Feature Layer Alignment*: The feature layer alignment mechanism breaks through the limitation of traditional distillation that only focuses on the output layer, and realizes deep knowledge migration through multi-level feature mapping. Aiming at the differences in feature dimensions of the intermediate layer of the teacher-student model, the spindle-shaped MLP adapter is invoked to carry out the dynamic dimensionality transformation at, and its loss function is defined as:

$$L_{spa} = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W \left\| \phi_{spa}(f_{tea}^{spa})_{ij} - \phi_{spa}(f_{stu}^{spa})_{ij} \right\|_2^2 \quad (21)$$

where ϕ_{spa} is the feature adapter and HW is the feature map dimension. The time-domain feature alignment module focuses on the temporal features of the ViT output, and realizes the feature space mapping through a learnable projection matrix:

$$L_{tmp} = \frac{1}{T} \sum_{t=1}^T \left\| W_{tmp} f_{tea,t}^{tmp} - W_{tmp} f_{stu,t}^{tmp} \right\|_2^2 \quad (22)$$

where $W_{tmp} \in \mathbb{R}^{64 \times d}$ is the dimension transformation matrix and T denotes the timing length. Frequency domain feature alignment combines wavelet transform and neural network, firstly, DWT is performed on the input signal to obtain four subbands, LL, LH, HL, and HH, and then frequency domain features are aligned by the adapter ϕ_{fre} :

$$L_{fre} = \left\| \phi_{fre}(DWT(f_{tea})) - \phi_{fre}(DWT(f_{stu})) \right\|_F^2 \quad (23)$$

The highest level of fusion feature alignment then directly matches the comprehensive characterization generated by the dynamic gating network:

$$L_{fis} = \frac{1}{256} \left\| \phi_{fis}(g_{tea}) - \phi_{fis}(g_{stu}) \right\|_2^2 \quad (24)$$

The four alignment paths are dynamically balanced by adaptive weights $w_k(t)$, which effectively solves the key problems of feature dimension mismatch and semantic information loss by establishing a complete hierarchical distillation system from local perception to global understanding.

3) *Total Loss Function*: In order to dynamically balance the importance of different distillation objectives, we propose an adaptive loss function based on learnable weights. A vector of trainable weight parameters is first introduced $\theta \in \mathbb{R}^4$, which generates a normalized weight distribution by softmax function:

$$w_k = \frac{\exp(\theta_k)}{\sum_{j=1}^4 \exp(\theta_j)}, \quad k = 1, 2, 3, 4 \quad (25)$$

where the four weight components correspond to the loss terms of spatial features w_{spa} , temporal domain features w_{tmp} , frequency domain features w_{fre} and fusion features w_{fus} . Based on the generated weights, the contribution of each feature alignment term is dynamically adjusted to form a weighted feature loss:

$$L_{Feat} = \sum_{k \in \{spa, tmp, fre, fis\}} w_k \cdot L_{MSE}^{(k)} \quad (26)$$

each feature loss term $L_{MSE}^{(k)}$ calculates the mean square error of the teacher and student models on the corresponding feature space. This dynamic weighting mechanism exhibits adaptive characteristics for attack types, enabling the model to automatically focus on the most relevant feature dimensions for different attack modes. The final total loss function balances the three types of critical losses through the hyperparameter α (set to 0.5 by default):

$$L_{Total} = \alpha L_{CE} + (1 - \alpha) L_{KL} + \beta L_{Feat} \quad (27)$$

above equation achieves the organic integration of three types of key loss functions: the cross-entropy classification loss L_{CE} ensures that the student model maintains its base recognition capability, the temperature-scaled KL dispersion distillation loss L_{KL} is responsible for migrating the knowledge of the teacher model's decision boundaries, and the weighted feature loss L_{Feat} ($\beta = 1$) improves the generalization performance of the student model by aligning the multilevel representations. These three are dynamically balanced by the hyperparameter α , and the network parameters and trainable weight parameter θ are synchronously optimized during back propagation to form a training mechanism with adaptive properties.

IV. EXPERIMENTS AND EVALUATION

A. Experimental Setting

1) *Experimental Environment*: The simulation experiments are Implemented on a device with Intel Xeon (R) W-2255 GHz, NVIDIA GeForce RTX 3090 and 128 GB RAM, which equipped with Ubuntu 20.04.1 LTS, PyCharm 2021.1.2 $\times$ 64, Python 3.8, and Tensorflow 2.10.0. The GPU is NVIDIA GeForce RTX 2080 Ti 11 GB.

2) *Datasets*: Our experiments utilized three benchmark datasets: Kitsune [30], CIRA-CIC-DoHBrw2020 [31], and CICIoT2023 [32], which include novel network attacks such as backdoors and worms, thereby reflecting current network intrusion behavior patterns. All datasets were split into training and testing sets at an approximate ratio of 8:2. Further details of each dataset are briefly described below.

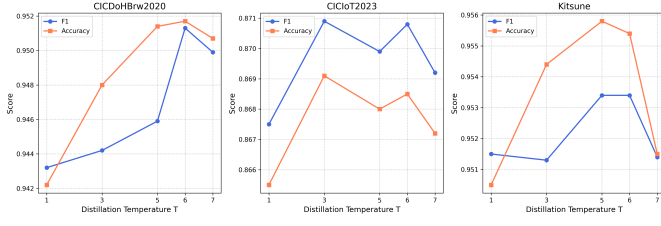


Fig. 4. Effect of distillation temperature T on F1 score and Accuracy across three datasets.

- Kitsune dataset contains traffic from a real IoT network testbed. The classification task is a 10-class problem, involving the detection of normal traffic and 9 distinct attack types: Video Injection, ARP MitM, Active Wiretap, SSDP Flood, SSL Renegotiation, Fuzzing, OS Scan, SYN DoS, and Mirai botnet.
- CIRA-CIC-DoHBrw2020 dataset focuses on encrypted DNS-over-HTTPS (DoH) traffic. The task is to identify benign traffic versus malicious activity conducted over DoH tunnels. The malicious traffic encompasses a wide range of threats, including Malware, Botnet, Data Exfiltration, Cryptojacking, Downloaded Attack, Phishing, Tor Connection, and DGA (Domain Generation Algorithm).
- CICIOT2023 is a large-scale dataset encompassing a wide variety of IoT traffic and attacks. The task is a comprehensive multi-class classification involving normal traffic and 33 specific attack types. These attacks are grouped into 7 major categories: DDoS, DoS, Mirai, Spoofing, Web-based, Reconnaissance, and Forgery.

3) *Evaluation Metrics*: The performance of the intrusion detection models is evaluated using multiple criteria. First, we assess detection accuracy through standard metrics: accuracy, precision, recall, and the F1-score. Second, model efficiency is quantified by the number of parameters and the model size. Finally, given the high security requirements of intrusion detection tasks, False Positive Rate (FPR) and False Negative Rate (FNR) are also provided. A high FPR, indicating frequent false alarms, can undermine operational efficiency and analyst trust. A high FNR, representing missed attacks, constitutes a severe security vulnerability. Reporting these metrics provides a balanced view of both the model's performance and its practical utility.

B. Hyperparameter Setting

1) *Distillation Temperature*: Knowledge distillation temperature critically determines the softening degree of teacher outputs. As shown in Figure 4, we evaluate $T \in \{1, 3, 5, 6, 7\}$ on all datasets. Results reveal dataset-dependent optimal temperatures. CICDoHBrw2020 achieves peak performance (F1 = 0.9513, Acc = 0.9517) at $T = 6$; CICIOT2023 attains its best F1 (0.8709) and accuracy (0.8691) at $T = 3$; and Kitsune yields the highest scores (F1 = 0.9534, Acc = 0.9558) at $T = 5$. Considering the need for a generalizable and stable temperature across diverse traffic types, we select $T = 5$ as the default setting. This choice is motivated by its near-optimal performance on all datasets (consistently within 2%

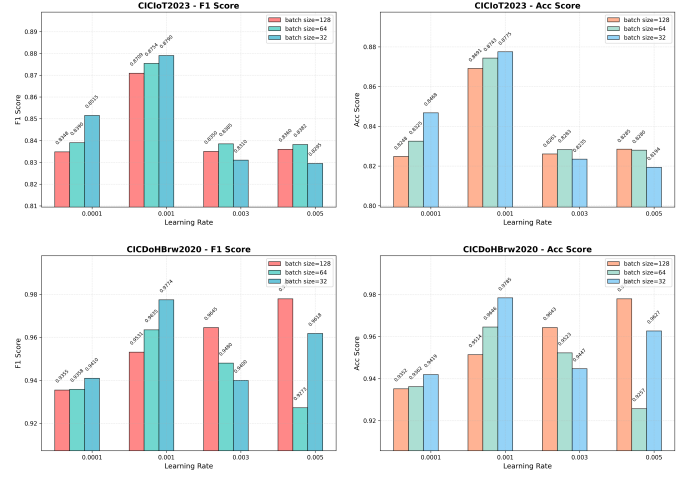


Fig. 5. Sensitivity analysis of learning rate and batch size on student model performance using grid search on CICIOT2023 and CICDoHBrw2020.

of the dataset-specific optimum) and its role in providing moderate softening—avoiding both overly sharp ($T = 1$) and over-smoothed ($T \geq 6$) distributions—thereby ensuring robust knowledge transfer without compromising task-specific discriminability.

2) *Learning Rate and Batch Size*: We conduct a joint grid search over learning rate and batch size (Figure 5). On CICIOT2023, $lr = 0.001$ combined with $batch_size = 32$ delivers the highest student F1 (0.8790) and accuracy (0.8775). On CICDoHBrw2020, $lr = 0.005$ together with $batch_size = 128$ yields the best result (F1 = 0.9779, accuracy = 0.9780). The sensitivity analysis reveals that smaller batch sizes generally yield better student performance but increase training time, while larger batches offer superior hardware utilization. Balancing computational efficiency, stable convergence, and cross-dataset robustness, we adopt $lr = 0.001$ and $batch_size = 128$ as the default configuration. This combination ensures reliable convergence behavior while maintaining GPU memory efficiency and consistent performance across different traffic profiles.

C. Traffic Classification Task

In order to comprehensively evaluate the performance of our proposed HKD-Net framework, we conducted extensive experiments on three public datasets: Kitsune, CIRA-CIC-DoHBrw2020, and CICIOT2023, and compared them with several state-of-the-art network intrusion detection methods, including DPCNN, DL-BiLSTM, TFTKD, CNN-LSTM, and CNN-GRU. These methods cover everything from convolutional neural networks and recurrent neural networks to hybrid models and advanced methods based on knowledge distillation.

DPCNN [33] propose a low-complexity text categorization framework based on pyramidal downsampling convolutional networks and residual concatenation, which dynamically captures remote semantic associations of text through hierarchical feature pyramids. DL-BiLSTM [34], combines Deep Neural Networks (DNNs) and Bidirectional Long Short-Term Memory Networks (BiLSTMs) for efficient intrusion detection

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THREE DATASETS

Approaches	Kitsune					CIRA-CIC-DoHBrw2020					CICIoT2023				
	Acc	Prec	Rec	FPR	FNR	Acc	Prec	Rec	FPR	FNR	Acc	Prec	Rec	FPR	FNR
DPCNN [33]	92.66	93.37	92.64	1.45	7.36	93.91	93.58	93.71	1.20	6.29	83.84	83.17	83.36	3.15	16.64
CNN-GRU [36]	96.71	96.82	96.76	0.47	3.31	95.34	95.75	95.71	0.18	1.24	83.74	85.96	83.86	1.90	12.89
CNN-LSTM [35]	96.54	96.71	96.52	0.64	4.48	94.98	95.18	94.97	0.86	5.03	83.37	84.85	83.30	2.46	16.58
TFTKD [10]	86.57	84.61	86.57	7.26	50.80	80.73	84.52	80.73	20.81	22.76	74.08	74.73	74.08	8.00	52.56
BILSTM [34]	96.33	96.51	96.16	0.48	3.38	95.65	95.69	95.64	0.22	1.65	84.96	85.08	84.66	1.86	11.10
HKD-Net	96.72	96.86	96.71	0.66	4.59	97.19	97.20	97.19	0.68	4.86	87.19	87.44	87.19	1.86	13.14

on resource-constrained IoT devices through feature down-sampling and model quantization techniques. TFTKD [10] propose a lightweight intrusion detection model based on two-dimensional Fourier transform and knowledge distillation, aiming at solving the problems of data sparsity, model complexity, and deployment efficiency in traditional intrusion detection. CNN-LSTM-Trans [35], A parallel detection framework based on hybrid Convolutional Neural Network (CNN) and Long Short-Term Memory Network (LSTM) is proposed to dynamically fuse spatio-temporal features through Transformer's self-attention mechanism. CNN-GRU [36], A parallel integration framework based on hybrid convolutional neural network (CNN) and gated recurrent unit (GRU) is proposed to dynamically optimize the spatio-temporal feature representations through a multi-domain feature fusion mechanism. Table I shows the performance metrics of HKD-Net and baseline methods on three datasets, including accuracy, precision, recall, FPR and FNR, which are widely used to evaluate the effectiveness of intrusion detection systems.

As shown in Table I, HKD-Net not only achieves leading accuracy, but also maintains a well-balanced performance in terms of both false positive rate (FPR) and false negative rate (FNR). On the CIRA-CIC-DoHBrw2020 dataset, HKD-Net attains the lowest FNR (2.81%) while maintaining a low FPR (0.55%), demonstrating its ability to minimize missed attacks without significantly increasing false alarms. On the more challenging CICIoT2023 dataset, HKD-Net achieves an accuracy of 87.19%, which is approximately 2 percentage points higher than the suboptimal DL-BiLSTM. Moreover, it attains the lowest FPR (1.86%) among all compared methods, while also delivering a low FNR (13.14%). These results indicate that our model offers distinct advantages in reducing false positives for normal traffic while preserving high detection sensitivity in complex multi-class scenarios.

Collectively, these results demonstrate the robustness and effectiveness of HKD-Net in detecting network intrusions across datasets of varying scales and protocol types.

D. Ablation Experiments

In this section, ablation studies are conducted to validate the effectiveness of the key components of HKD-Net.

1) *Multi-domain Feature Fusion*: In this section, we compare the performance of five different models on the Kitsune, DoHBrw2020, and CICIoT2023 datasets, respectively: the full model (HKD-Net), the model with the time-domain feature extraction module removed (HKD_VIT), the model

TABLE II
RESULTS OF THE EVALUATION OF THE EFFECTIVENESS OF MULTI-DOMAIN FEATURE FUSION

Datasets	Methods	Acc	Precision	Recall	F1
CICIoT2023	HKD-Net	0.8719	0.8744	0.8719	0.8732
	HKD_SE	0.8569	0.8620	0.8511	0.8565
	HKD_VIT	0.8672	0.8717	0.8671	0.8694
	HKD_WT	0.8511	0.8672	0.8515	0.8593
	HKD_DFF	0.8664	0.8724	0.8626	0.8674
DoHBrw2020	HKD-Net	0.9719	0.9720	0.9719	0.9719
	HKD_SE	0.9538	0.9538	0.9537	0.9538
	HKD_VIT	0.9516	0.9514	0.9516	0.9515
	HKD_WT	0.9597	0.9597	0.9597	0.9597
	HKD_DFF	0.9299	0.9318	0.9299	0.9309
Kitsune	HKD-Net	0.9672	0.9686	0.9671	0.9674
	HKD_SE	0.9580	0.9586	0.9580	0.9583
	HKD_VIT	0.9489	0.9491	0.9489	0.9490
	HKD_WT	0.9643	0.9667	0.9643	0.9655
	HKD_DFF	0.9574	0.9539	0.9511	0.9506

with the frequency-domain feature extraction module removed (HKD_WT), the model with the spatial domain feature extraction module removed (HKD_SE), and the model with the dynamic feature fusion mechanism removed (HKD_DFF). These experiments aim to quantify the specific contribution of each part to the overall performance by removing the key components of the model one by one.

The results are summarized in Table II. Overall, HKD-Net demonstrates the highest and most consistent performance. Ablation studies reveal that each feature domain contributes significantly to model effectiveness. The removal of the spatial, temporal, and frequency modules leads to accuracy drops of approximately 2.0%, 1.5%, and 1.0% respectively, with consistent trends observed across datasets. Furthermore, removing the dynamic feature fusion mechanism decreases the performance metrics by 0.55%, 0.5%, 0.5% and 0.5%, respectively. These findings emphasize the significant impact of the lack of a dynamic feature fusion module in feature fusion, suggesting that multi-domain feature fusion needs to be nuanced for a specific context, allowing for varying degrees of involvement of features from different domains.

2) *Multi-Level Knowledge Distillation*: In order to systematically evaluate the contribution of each component in hierarchical distillation, this paper designs comparison experiments of three distillation strategies on three publicly available datasets, namely Kitsune, CIRA-CIC-DoHBrw2020 & CICIoT2023. The experiments conduct ablation experiments by taking the complete HKD-Net (with both output-layer KL distillation and intermediate-layer feature alignment) as the

TABLE III
EFFECTIVENESS EVALUATION RESULTS OF DIFFERENT
DISTILLATION STRATEGIES

Dataset	Model	Acc (%)	Precision	Recall	F1
Kitsune	HKD-Net	96.72	96.86	95.71	96.74
	KD_CE	93.06	93.39	93.39	93.51
	KD_KL	88.25	89.94	88.25	87.96
DoHBrw2020	HKD-Net	97.19	97.20	97.19	97.19
	KD_CE	94.36	94.51	94.36	94.48
	KD_KL	96.75	96.77	96.75	96.77
CICIoT2023	HKD-Net	87.19	87.44	87.19	87.32
	KD_CE	84.90	84.49	84.16	84.32
	KD_KL	85.71	85.50	85.52	85.32

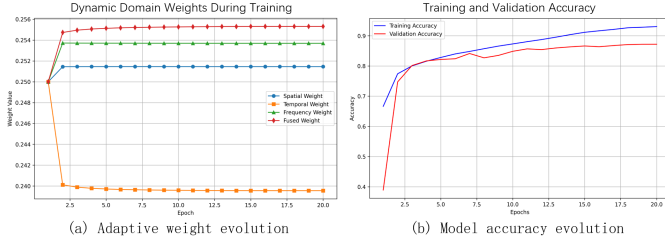


Fig. 6. Dynamic evolution of adaptive weights (spatial, temporal, frequency, fusion) and validation accuracy during training on the Kitsune dataset.

baseline, and evaluating three variants: KD_CE (hard-label training only), KD_KL (output-layer distillation only), and the intermediate-feature-alignment-only model.

As shown in Table III, the complete HKD-Net framework achieves the best performance across all datasets. Compared to using only hard labels (KD_CE) or output-layer distillation (KD_KL), hierarchical distillation provides significant improvements. For instance, on Kitsune, it surpasses KD_KL by 8.78% in F1 score. Even on encrypted traffic scenario (DoHBrw2020) and complex (CICIoT2023) traffic, it maintains clear advantages. Overall, output-layer and intermediate feature alignment are complementary, with the former ensuring semantic consistency and the latter capturing fine-grained representations, together establishing the effectiveness of hierarchical distillation.

3) *Adaptive Knowledge Distillation*: In this paper, an adaptive loss function based on learnable weights is adopted to guide the lightweight student model to automatically focus on the most discriminative modal representations under dynamic data distributions by evaluating the quality of the alignment of each modal feature with the teacher model in real-time during the training process and adaptively adjusting the relative weights of spatial, time-domain, frequency-domain, and fusion features in the distillation loss. Compared to the traditional fixed-weight scheme, this strategy simplifies the engineering process by improving accuracy without the need for additional hyperparameter tuning.

Figure 6 shows the dynamic evolution of adaptive weights across the spatial, temporal, frequency, and fusion domains during training (taking the Kitsune dataset as example). The weights exhibit clear differentiation in the early training phase, with the validation accuracy curve rising sharply in tandem. This concurrent behavior confirms the efficacy of the adaptive weighting mechanism.

To quantify its advantage, we compare the Adaptive Weighting (AW) strategy against a Fixed Weighting (FW) baseline under controlled conditions (see Table IV). Except for the weighting strategy, the teacher model, the student model architecture, the number of training rounds, the optimizer, and the random seed are all kept the same. The AW strategy achieves significantly higher validation accuracy (+ 1.34%), reduces inference latency by 14.2 ms, and converges 2 epochs earlier than the FW approach, with all improvements statistically significant ($p < 0.01$). Especially, the AW mechanism accelerates convergence speed by approximately 16% while substantially reducing inference time, highlighting its practical effectiveness in expediting model training. Furthermore, the low standard deviations ($< 0.1\%$) across three independent runs confirm the robustness of the results.

In summary, learnable adaptive weights simultaneously achieve higher accuracy, faster convergence and lower inference latency without increasing the number of parameters and computational overhead, providing a concise and effective solution for cross-modal knowledge distillation.

E. Edge Deployment Feasibility Analysis

1) *Resource Requirements and Compatibility*: As shown in Table V, the student model of HKD-Net occupies only 1.06MB of storage and 4.24MB of peak runtime memory. This corresponds to minimal resource utilization—e.g., 0.003% storage and 0.10% memory on a Raspberry Pi 4. Even on simulated low-end devices (1GB RAM), memory usage remains below 0.5%. Full compatibility across all tested platforms confirms that HKD-Net’s footprint fits well within typical edge-hardware constraints, enabling practical deployment in resource-limited environments.

2) *Computational Efficiency and Performance Trade-offs*: As shown in Table VI, HKD-Net student achieves a remarkable balance between performance and efficiency. With only 0.0085 GFLOPs, it is $2.4\times$ more computationally demanding than the lightest baseline TFTKD (0.0035 GFLOPs), yet delivers significantly superior F1 scores across all datasets. Compared to its teacher (0.0634 GFLOPs), the student reduces FLOPs by 86.6% while retaining over 97% of its F1 score. Furthermore, HKD-Net achieves inference latencies of 95–101ms. These confirm its practical viability for resource-constrained edge environments where detection accuracy and operational efficiency must be jointly satisfied.

F. Adversarial Robustness Via Cross-Dataset Evaluation

We consider the scenario where adversarial traffic manifests as novel attack patterns or modified traffic signatures that induce distribution shifts relative to the training data. To evaluate robustness under such realistic adversarial conditions, we trained HKD-Net on the CICDoHBrw2020 dataset and tested its performance on the original test set (baseline) and the CICIoT2023 dataset—which contains multiple attack categories not present during training (e.g., DDoS, Web-based, Spoofing). To ensure fair comparison, we converted multi-class tasks to binary classification (normal vs. adversarial) using dataset-specific normal class identification.

TABLE IV
QUANTITATIVE COMPARISON OF PAIRED T-TESTS WITH ADAPTIVE WEIGHTS AND FIXED WEIGHTS

Indicator	AW 42	AW 43	AW 44	FW 42	FW 43	FW 44	Delta Mean	Paired t-test
Accuracy	0.9672	0.9685	0.9669	0.9541	0.9535	0.9547	+1.34%	$t(2)=15.6, p_1 0.01$
Inference Time (ms)	83.7	84.0	83.8	97.9	98.1	98.0	-14.2	$t(2)=38.4, p_1 0.001$
Acc convergence epoch	10.5	10.3	10.6	12.5	12.4	12.6	-2.0	$t(2)=-21.2, p_1 0.01$

TABLE V
HKD-NET EDGE-DEPLOYMENT RESOURCE REQUIREMENTS

Device	Storage Cap.	Model Size	Storage Util.	Memory Cap.	Est. Memory	Memory Util.
Raspberry Pi 4	32 GB	1.06 MB	0.003%	4 GB	4.24 MB	0.10%
Jetson Nano	16 GB	1.06 MB	0.007%	4 GB	4.24 MB	0.10%
Simulate low-end devices	4 GB	1.06 MB	0.03%	1 GB	4.24 MB	0.41%

TABLE VI
COMPUTATIONAL COMPLEXITY COMPARISON OF DIFFERENT METHODS ON THREE DATASETS

Approaches	Overall		Kitsune		CIRA-CIC-DoHBrw2020		CICIoT2023	
	Parameters/M	FLOPs/G	Inference Time/ms	F1	Inference Time/ms	F1	Inference Time/ms	F1
DPCNN [33]	0.5135	0.0358	73.50	0.9280	65.26	0.9351	66.50	0.8204
CNN-GRU [36]	0.2155	0.0239	72.49	0.9678	68.23	0.9554	69.31	0.8484
CNN-LSTM [35]	0.0452	0.0082	75.17	0.9618	62.42	0.9506	66.89	0.8497
TFTKD [10]	0.0085	0.0024	28.65	0.8612	29.97	0.8258	29.17	0.7740
BILSTM [34]	0.2339	0.0152	72.67	0.9633	60.68	0.9392	62.12	0.8434
HKD-Net(Teacher)	1.1002	0.0634	126.76	0.9782	123.13	0.9841	123.81	0.8894
HKD-Net(Student)	0.2786	0.0085	100.74	0.9674	94.82	0.9719	96.50	0.8732

TABLE VII
CROSS-DATASET ROBUSTNESS EVALUATION OF
HKD-NET UNDER DISTRIBUTION SHIFT

Dataset	Accuracy (%)	F1-Score	Detection Rate
DoHBrw2020 (Baseline)	95.05	95.07	89.16
DoHBrw2020 (Cross)	99.92	99.92	99.98
CICIoT2023 (Cross)	86.70	81.95	99.00

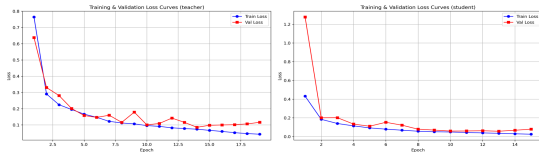


Fig. 7. Comparison of training/validation loss evolution between the teacher model and the student model.

Table VII presents the cross-dataset evaluation results. The student model maintains strong detection capability on unseen attack types, achieving 99.00% detection rate on CICIoT2023 (containing DDoS, Web attacks, etc. not present in training). Notably, on the same-distribution CICDoHBrw2020 test set, the model achieves 99.92% accuracy and 99.98% detection rate, outperforming its original performance. These results validate that knowledge distillation effectively enhances the student model's generalization to unseen traffic variations. Crucially, achieving a 99.00% detection rate on entirely absent attack patterns demonstrates HKD-Net's robustness against adversarial traffic, even under significant distribution shift from known attacks.

G. Training Process and Knowledge Migration Analysis

In order to thoroughly analyze the optimization mechanism and knowledge transfer effect of the HKD-Net model, this

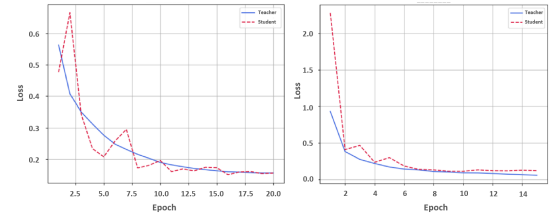


Fig. 8. Loss curves of the student model under two strategies: output layer distillation and complete distillation (including alignment of output layer and intermediate layer features).

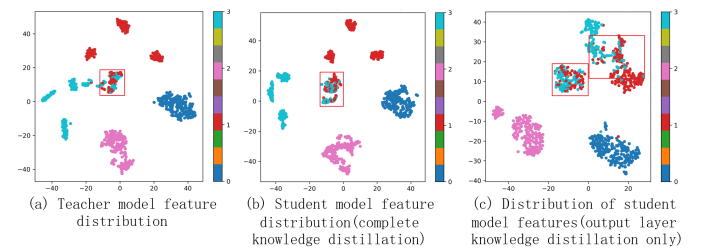


Fig. 9. t-SNE visualization of the distribution of teacher and student model features.

section conducts a systematic study from three dimensions: loss function convergence characteristics, feature space evolution, and dynamic weight analysis.

1) *Loss Function Convergence Characteristics*: As shown in Figure 7, the teacher model exhibits rapid and stable convergence, while the student trained solely with output-layer distillation shows slower initial convergence and fluctuating validation loss. A further comparison in Figure 8 demonstrates that introducing intermediate feature alignment and adaptive weighting significantly accelerates the student model's

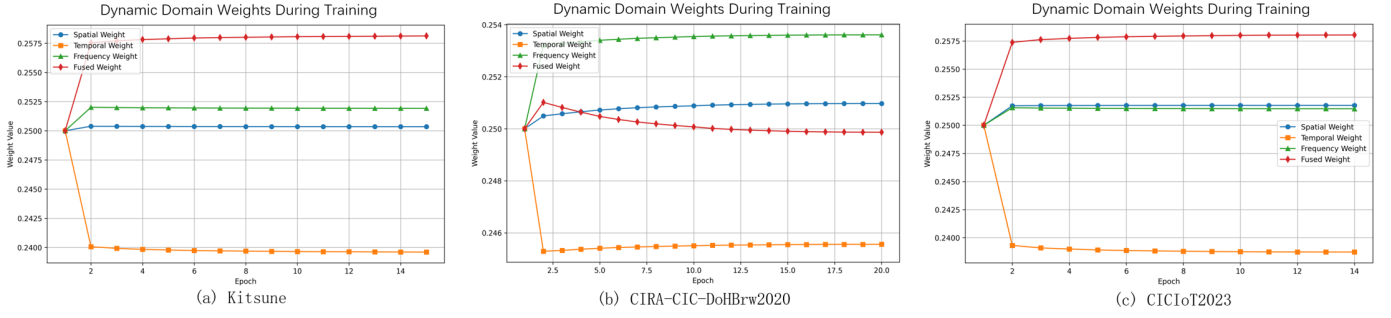


Fig. 10. Epoch change curves of dynamic weights on various datasets.

convergence and stabilizes its validation loss compared to using output-layer distillation alone. These results confirm that our hierarchical distillation strategy effectively facilitates knowledge transfer and enables the student model to better approximate the teacher’s performance.

2) *Feature Space Evolution*: We use the t-SNE algorithm to visualize the features of the key layers to present the differences and evolutionary process between the teacher model and the student model in the feature space. The results are shown in Figure 9. The teacher model exhibits compact, well-separated clusters across spatial, temporal, and frequency domains. In contrast, the student trained with only output distillation shows dispersed and overlapping distributions. Under full hierarchical distillation, the student’s features progressively align with the teacher’s, forming more discriminative clusters. This demonstrates that intermediate feature alignment and adaptive weighting effectively guide the student to learn richer and more structured representations, contributing to improved detection performance.

3) *Dynamic Weight Analysis*: As depicted in Figure 10, the adaptive weights for spatial, temporal, frequency, and fusion domains evolve distinctly across datasets, reflecting domain-specific feature importance. For instance, frequency weights rise rapidly on CICDoHBrw2020, while fusion weights dominate early training on Kitsune and CICIoT2023. Weights gradually stabilize as training proceeds, with each dataset reaching a distinct equilibrium. This demonstrates the mechanism’s ability to dynamically recalibrate modal contributions based on dataset characteristics, thereby optimizing fusion performance. Moreover, modalities assigned higher weights correlate strongly with improved detection performance, confirming that the weighting strategy successfully prioritizes the most discriminative features per scenario.

V. DISCUSSION

This study aims to address the effectiveness of balancing the detection accuracy and computational efficiency of network intrusion detection on resource constrained devices. Unlike conventional knowledge distillation methods, our framework introduces a hierarchical distillation strategy that explicitly aligns intermediate-level feature representations across temporal, spatial, and frequency domains. Additionally, by leveraging gated networks to adaptively assign weights to the contributions of various feature domains, we can construct

representations that are more context-aware and discriminative. Experimental results validated the effectiveness of our approach in various datasets, including Kitsune, CIRA-CIC-DoHBrw2020, and CICIoT2023. The student model with only 0.278M parameters and 0.0064 GFLOPs, achieves comparable accuracy to the teacher and consistently outperforms lightweight baselines like TFTKD across all datasets (e.g., +10.6% in F1-score on Kitsune). This design enables the lightweight student model to inherit the teacher’s comprehensive representational capacity across multiple domains.

One potential limitation of our research is that the robustness of HKD-Net to novel zero day attacks has not been validated, as the model was trained and tested on known attack types. Second, although we have conducted resource feasibility analysis on simulated edge devices (as shown in Table V), the actual deployment of HKD-Net on physical edge hardware under sustained operational load requires further validation. Real-world deployment may encounter constraints not captured in our server environment, including memory bandwidth limitations and variable power budgets. In the future, we plan to integrate an online incremental learning framework into HKD-Net, enabling the model to continuously adapt to emerging attack patterns without costly full retraining. Meanwhile, we will implement HKD-Net on physical edge devices to conduct rigorous benchmarking of its real-time inference latency, power consumption, and long-term stability under realistic network conditions.

VI. CONCLUSION

In this paper, we propose HKD-Net, a hierarchical knowledge distillation framework for efficient network intrusion detection on resource-constrained edge devices. By integrating temporal, spatial, and frequency domain feature extraction with a dynamic gating mechanism, HKD-Net constructs a discriminative multi-domain representation. Through hierarchical distillation that aligns both output logits and intermediate features, the lightweight student model achieves performance comparable to the teacher while reducing parameters by 74.68% and maintaining low computational cost. Experimental results on three public datasets demonstrate that HKD-Net outperforms state-of-the-art methods in accuracy and F1-score while significantly lowering false positive and false negative rates. The model also exhibits efficient resource usage, confirming its suitability for edge deployment. In future work, we will focus on real-world edge validation and extend

HKD-Net towards open-world scenarios, including robustness testing against zero-day attacks and the integration of incremental learning for adaptive threat detection.

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